
This exam has 2 pages and 7 exercises.
The result will be computed as (total number of points plus 10) divided by 10.

1. Sheffer stroke (9 points)

Is the Sheffer stroke | associative? Show this by proving the semantic equivalence $(\phi \mid \psi) \mid \chi \equiv \phi \mid (\psi \mid \chi)$, or bring a counterexample to disprove it.

2. Island puzzle (12 points)

On the island of liars and truth speakers, every resident either always speaks the truth or always lies. Suppose that you meet three islanders, A, B and C.

> A says *'B is a liar'*,
> B says *'C is a truth speaker'*, and
> C says *'A and B are both liars'*.

Determine, via a truth table, whether A, B and C are liars or truth speakers.

3. CNF algorithm (12 points)

Show, using the CNF algorithm, that the formula $(p \wedge \neg q) \rightarrow \neg(q \rightarrow p)$ is semantically equivalent to $\neg p \vee q$.

4. DPLL procedure (12 points)

Apply the DPLL procedure to the CNF $(p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg q \vee r)$ to show whether it is satisfiable. Explain in detail how the procedure comes to a result. Unless there is unit clause or a pure literal, let the procedure branch first on *p*, then on *q*, and then on *r*; and let it always try the truth value true (T) before false (F).

5. Sets (6 + 9 points)

Consider the following set-theoretic equality:

$$(A' \setminus B) \cap (C \setminus B) = (A \cup C')' \setminus B$$

(a) Prove this identity using a sequence of Venn diagrams. For each side of the equality, you can draw a sequence of diagrams which shows how the set is constructed from A, B, and C via elementary set operations, and use this to prove that the equation holds for any three sets A, B and C.

(b) Prove this identity using the rules of the algebra of sets. Please make sure to name the rules which you apply.

6. Relations (5 + 6 + 6 points)

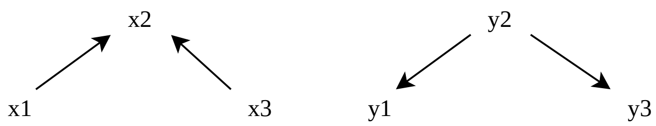
Consider the set V = {a, b, c, d, e} and W = {1, 2, 3, 4, 5} and the relations:

$$R = \{(a, a), (a, d), (b, b), (c, c), (c, d), (d, a), (d, c), (d, e), (e, e)\}$$
$$S = \{(a, 2), (b, 3), (d, 3), (d, 5), (e, 4)\}$$

a) Draw the directed graph representation of the relation R.
b) Explain whether the relation R is transitive and/or anti-symmetric.
c) Write down the matrix representation of the composite relation $S \circ R$. It might help to illustrate the relations S, R and the composition $S \circ R$ in a Venn-diagram first. Do not use matrix multiplication!

7. Ordering relations (7 + 4 points)

Consider the sets $X \coloneqq \{x_1,x_2,x_3\}$ and $Y \coloneqq \{y_1,y_2,y_3\}$, with partial orders $\leq_X$ and $\leq_Y$ on these respective sets described by the following Hasse diagrams:



(a) Draw the Hasse diagram of the Cartesian ordering relation on $X \times Y$ induced by $\leq_X$ and $\leq_Y$ by adding arrows to the picture below. You can use the algorithm you have learned about in class, but you are not required to do so.

$\langle x_1, y_1\rangle$	$\langle x_1, y_2\rangle$	$\langle x_1, y_3\rangle$
$\langle x_2, y_1\rangle$	$\langle x_2, y_2\rangle$	$\langle x_2, y_3\rangle$
$\langle x_3, y_1\rangle$	$\langle x_3, y_2\rangle$	$\langle x_3, y_3\rangle$

(b) Determine and list all the maximal and all the minimal elements of $X \times Y$ in this Cartesian ordering. Is there a maximum? Is there a minimum?

End of the exam.